Now let us recall the properties of the real number system which is the foundation for mathematics.

2.2.4 Properties of Real Numbers

(i) For any $a, b \in \mathbb{R}$, $a + b \in \mathbb{R}$ and $ab \in \mathbb{R}$.
[Sum of two real numbers is again a real number and multiplication of two real numbers is again a real number.]

(ii) For any $a, b, c \in \mathbb{R}$, $(a + b) + c = a + (b + c)$ and $a(bc) = (ab)c$.
[While adding (or multiplying) finite number of real numbers, we can add (or multiply) by grouping them in any order.]

(iii) For all $a \in \mathbb{R}$, $a + 0 = a$ and $a(1) = a$.

(iv) For every $a \in \mathbb{R}$, $a + (-a) = 0$ and for every $b \in \mathbb{R} - \{0\}$, $b\left(\frac{1}{b}\right) = 1$.

(v) For any $a, b \in \mathbb{R}$, $a + b = b + a$ and $ab = ba$.

(vi) For $a, b, c \in \mathbb{R}$, $a(b + c) = ab + ac$.

(vii) For $a, b \in \mathbb{R}$, $a < b$ if and only if $b - a > 0$.

(viii) For any $a \in \mathbb{R}$, $a^2 \geq 0$.

(ix) For any $a, b \in \mathbb{R}$, only one of the following holds: $a = b$ or $a < b$ or $a > b$.

(x) If $a, b \in \mathbb{R}$ and $a < b$, then $a + c < b + c$ for all $c \in \mathbb{R}$.

(xi) If $a, b \in \mathbb{R}$ and $a < b$, then $ax < bx$ for all $x > 0$.

(xii) If $a, b \in \mathbb{R}$ and $a < b$, then $ay > by$ for all $y < 0$.



Exercise - 2.1

1. Classify each element of $\{\sqrt{7}, \frac{-1}{4}, 0, 3.14, 4, \frac{22}{7}\}$ as a member of $\mathbb{N}$, $\mathbb{Q}$, $\mathbb{R} - \mathbb{Q}$ or $\mathbb{Z}$.
2. Prove that $\sqrt{3}$ is an irrational number.
(Hint: Follow the method that we have used to prove $\sqrt{2} \notin \mathbb{Q}$.)
3. Are there two distinct irrational numbers such that their difference is a rational number? Justify.
4. Find two irrational numbers such that their sum is a rational number. Can you find two irrational numbers whose product is a rational number.
5. Find a positive number smaller than $\dfrac{1}{2^{1000}}$. Justify.

2.3 Absolute Value

2.3.1 Definition and Properties

As we have observed that there is an order preserving one-to-one correspondence between elements of $\mathbb{R}$ and points on the number line. Note that for each $x \in \mathbb{R}$, x and $-x$ are equal distance from the origin. The distance of the number $a \in \mathbb{R}$ from 0 on the number line is called the *absolute value* of the number a and is denoted by $|a|$. Thus, for any $x \in \mathbb{R}$, we have

$$|x| = \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0. \end{cases}$$

and hence $|\cdot|$ defines a function known as absolute value function, from $\mathbb{R}$ onto $[0, \infty)$ and the graph of this function is discussed in Chapter 1.